

4

B

Let time interval between each drop be t .

First drop took $3t$ to reach the ground and travelled 9 m

$$9 = 0(4t) + \frac{1}{2}(g)(3t)^2$$

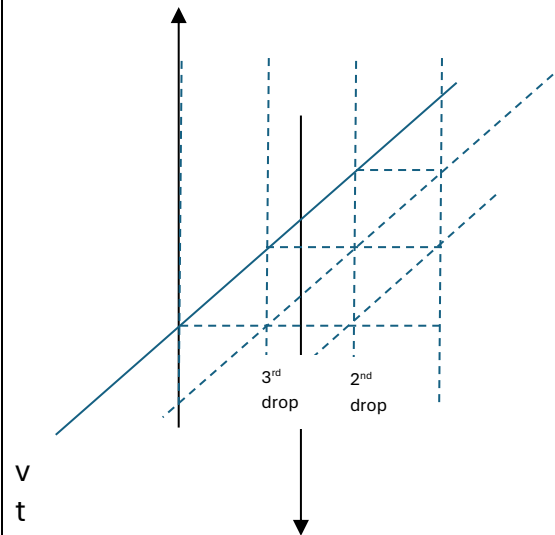
$$t^2 = \frac{2}{g}$$

distance travelled by 2nd drop = $0(2t) + \frac{1}{2}g(t)^2$

distance travelled by 3rd drop = $0(3t) + \frac{1}{2}g(2t)^2$

distance between 2nd and 3rd drop = $\frac{1}{2}g(2t)^2 - \frac{1}{2}g(t)^2 = 3.0\text{ m}$

alternative solution:



v

t

4th drop start to fall

1st drop touch the ground

distance between 1st drop and 4th drop = 9 triangles = 9 m

distance between 2nd and 3rd drop = 3 triangles = 3 m

5

A

$$E = 70 - (20+6)g$$